Deterioration states — methodology

Collections risk modelling · deterioration states · docs/deterioration_states/METHODOLOGY.md

How a credit-card account in buckets 4–7 fell into delinquency, and what it is doing now, read from its DPD bucket history only. One state per account, assigned by a short decision tree. MECE by construction, simple enough to explain to the risk team in one slide, and designed to stay out of the way of the other persona axes.

> **Status.** Designed and validated on calibrated **synthetic** data. Not yet run on the bank's book. Every number below marked *(synthetic)* is from simulation and is a *method check*, not a finding about the portfolio. The real-data run is `snippets/dpd_states_validate.py`; its expectations are in §8.

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1. Where this sits among the persona axes

Axis	Question it answers	Inputs
Locatability	Can we reach them?	Reachability flag, in-country flag
Ability	Can they pay?	Score: salary, inflow, CASA balance
Payment credibility	Do they pay and keep their word?	Score: promises kept (0–2), money coming in last 3 months (0–2), pre-delinquency history clean (0–1)
Deterioration (this)	**How did they fall in, and are they holding now?**	**DPD bucket history only**

Design rules agreed with the business owner:

1. **MECE** — every account gets exactly one state; no "other" bucket.
2. **Simplicity beats precision** — a rule a collector can repeat beats a better-fitting one nobody can explain. Thresholds are round numbers.
3. **Orthogonal where possible** — the states must not rebuild the credibility or ability score. Known overlaps are named (§7), not hidden.
4. **5–6 states, at most 8**, each with a different action for the risk team.

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2. Population and data conventions

- **Spine:** credit-card accounts whose **current** bucket is 4, 5, 6 or 7. Bucket 8+ is a separate category. Full history is available for every account.
- **The string:** one digit per month, **leftmost = current month** (`54320002000` → today 5, last month 4, …). Configurable if a source is reversed.
- **0 ≡ 1.** Bucket 1 (a few days late) is treated as normal.
- **Monthly movement `d` = bucket this month − bucket last month**, which reads as a payment:

d	Meaning	Months paid that month (= 1 − d)
+1	roll — paid nothing (or less than one month's due)	0
0 at bucket ≥ 2	**hold** — paid one month's due, stays in the same bucket	1

d	Meaning	Months paid that month (= 1 − d)
−k	paid k + 1 months, moved back k buckets	k + 1
back to 0/1	**cure** — common from 2–3, very rare from 4+	—

- **Spell:** a continuous run of months at bucket ≥ 2. The *current spell* is the one the account is in today.
- **Blip:** an earlier spell of 1–2 months that never passed bucket 2 (`0 0 2 0 0`). Blips do not count as earlier spells and do not end the clean history.
- **Billing-cycle jumps:** a +2 move next to a hold (`3 3 5` or `3 5 5`) is the reporting artefact of a straight roll and is repaired to `3 4 5` . Any other +2 is read as a roll.

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3. Features used by the states

Only features that **do not drift with time since reaching 4** are used to assign states. For most of the spine the current bucket is a clock (one bucket per month), so any count over a fixed window, or any rate over the whole spell, changes with the bucket rather than with behaviour.

Feature	Definition	Fixed once past 4?
`months_before_spell`	months of history before the current spell started	yes
`max_prior_bucket`	deepest bucket reached in any earlier (non-blip) spell; 0 if none	yes
`runup_len`	months from the last clean month to the first month at bucket 4 (3 = straight 2→3→4)	yes
`runup_pays`	months on the way to 4 in which 2+ months were paid (moved back)	yes
`last6_in_band`	of the last 6 months (incl. today), how many were within today's bucket ±1	no — current behaviour, by design

Other features (`n_blips` , `runup_holds` , `post4_paid_rate` , `max_hold_run` , …) are computed for diagnostics and discovery only.

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4. The states

Questions are asked **in order**; the first "yes" assigns the state.

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Q1  Bust-out?   <= 3 months of history before this spell
                (at most one blip inside it — a payoff from 2 — and no earlier real spell)
                AND run-up to 4 took <= 3 months                    -> bust_out
Q2  Holding?    >= 5 of the last 6 months within today's bucket ±1  -> holding
Q4  Relapse?    [B, C] an earlier spell reached bucket 3 and cured  -> relapse
Q5  Fell fast?  run-up (last clean month -> first month at 4) <= 3 months
      yes -> sudden    [B, C: >= 12 months of history before the spell
                        -> sudden_established, else sudden_early_life]
      no  -> gradual   [C: paid 2+ months at once on the way down
                        -> gradual_fought_back, else gradual_slide]
```

(Q3 "partial payer" was tested and rejected — §6.)

Three options

Option	States	Use when
A — minimal (4)	bust_out, holding, sudden_decline, gradual_decline	Strict orthogonality: prior-spell history is left entirely to credibility
B — balanced (6), recommended	bust_out, holding, relapse, sudden_established, sudden_early_life, gradual_decline	Default. Matches the natural groups in the data (§6.1) and the 5–6 target
C — detailed (7)	B with gradual split into gradual_slide / gradual_fought_back	The risk team has a different action for "slid" vs "fought back" and both are ≥ 4%

What each state means and the action it points to

State	Reads as	Action flavour	Share *(synthetic)*
bust_out	At most 3 months of history (one blip allowed), rolled straight in, never really paid	Fraud / first-party review, early legal; no restructuring	6% (bank: 6%)
holding	Paying about one month's due each month for 5 of 6 months	**Offer restructuring to break the hardship**	7% (bank's stabilized: 6%)
relapse	Has been here before and cured from bucket 3	Has shown they can catch up: firm promise-based plan; credibility decides the pressure	16%
sudden_established	At least a year of clean history (≥ 12 months), then a straight roll	Life event. Read with the other axes: not in country → departed; no salary → income shock; salary still in → won't pay. Most likely to respond to a hardship route	52%
sudden_early_life	4–11 months of clean history (first year on book), then a straight roll	Same triage; also an underwriting signal for acquisition	3% — below the 5% floor in synthetic data; the real share decides (E16)
gradual_decline	Paid something on the way down but kept slipping	Over-indebted: longer-tenor restructure or settlement	16%
↳ gradual_slide *(C)*	Paid one month here and there, never caught up	Chronic insufficiency: restructure to what they actually pay	10%
↳ gradual_fought_back *(C)*	Paid 2+ months at once, then rolled anyway	Lumpy / irregular income: plan aligned to inflow timing	6%

Holding is the only state that moves month to month — it describes what the customer is doing now (*synthetic:* 62% are still holding next month). Every other state is fixed once the account is past bucket 4 (*synthetic:* 97–100% keep their state next month).

5. Thresholds and the evidence behind them *(synthetic sweeps)*

Threshold	Value	Why this value	What moving it does
Bust-out history	**≤ 3 months, at most one blip**	Business definition (agreed): a customer who never pays misses the first or second bill, so 2–3 clean months at most (`0 0 2 3 4` , `0 0 0 2 3 4` , `0 2 0 2 3 4`). No natural knee in the data	*Synthetic:* at 3 months 93% of the state is a planted bust-out; at 6 months 79%, at 12 months 55% — longer histories are early-life defaults and land in sudden_early_life. Sweep reported in every run (2–12 months)
Bust-out earlier spells	**none; one blip allowed**	A bust-out may pay off once, but only from bucket 2 (a blip)	`0 2 3 4 0 0 2 3 4 5 6` is therefore relapse, not bust-out; two blips (`0 2 0 2 0 2 3 4`) is not a bust-out
Run-up for "sudden"	**≤ 3 months**	The one sharp threshold: any single paid month on the way down makes it gradual	≤ 4 moves ~70% of genuine slow slides into sudden (slides in gradual: 98% → ~30%)
Holding band	**5 of last 6 months within ±1**	The bank's existing stabilized rule, unchanged	4 of 6: share ~14%, next-month roll ~46%, persistence ~63%. 6 of 6: ~5%, ~32%, ~80%
Relapse depth	**earlier spell reached 3**	The bank counts ~85% single-spell; spells from 3 give 83% *(synthetic)*	From 2: 19%. From 4: 5% (cures from 4+ are rare)
Established	**≥ 12 months**	Business definition (agreed): the first year on book is the standard early-life default window. Bust-out already takes ≤ 3 months, so early life = 4–11 months	*Synthetic:* 12 m gives early life only 3.4% (below the 5% floor), 18 m 7%, 24 m 11%, 36 m 20% — but these shares come from the generator's assumed tenure distribution, so the real share decides (E16)

Rule order. Bust-out is asked before holding (the bank's order). Only ~0.2% of the spine is a bust-out that is also holding today, so the order barely matters.

6. What was tried and dropped — and why

6.1 Unsupervised clustering as the source of states (k-means, GMM)

- **Whole spine, k-means:** very stable (bootstrap ARI 0.99) but only ARI 0.3–0.4 against the planted patterns and 0.1–0.3 against the rule states. It always merged holders with staircases and never isolated slow slides. **Stability does not mean validity** — on real data stability is the only check available, and it cannot detect merged behaviours.
- **GMM:** BIC never reached a minimum (kept improving to 12 components) because the features are spiky — 56–93% of accounts sit on a single value (run-up = 3, no earlier spell, …), and GMM fits narrow components to the spikes. Under 1% of accounts were ambiguous, so the soft-assignment advantage disappears. Best ARI vs planted 0.48 but stability fell to 0.67–0.93. Not used. GMM fits continuous inputs (ability scores, balances) far better than DPD-derived counts.
- **Residual clustering (after bust-out and holding) — kept as a diagnostic.** k-means at k = 6 on the remaining ~88% found six clean, nameable groups: long-clean straight roll, short-clean straight roll, earlier spell to 3+, paying after 4 but still rolling, held on the way, paid 2+ on the way. With the final thresholds, agreement at k = 6 is A 0.11, **B 0.32, C 0.34** (best at k = 3: B 0.65 vs A 0.31) — the evidence

that two residual states were too few. Every B/C split maps to one of these clusters with a one-line rule; the clusters themselves are not the states. The clusters split clean history near 36 months, but that reflects the generator's assumed tenure distribution; the established cut is a business definition (12 months, §5).

6.2 A "mixed" / residual state

Dropped. An ordered tree with a default branch is MECE without one, and a residual bucket invites "everything we don't understand" to grow.

6.3 Staircase (holds for months at each level, then steps up)

Merged. While it is holding it *is* holding (that is the actionable fact); just after a step it reads as gradual. Telling a genuine holder from a staircase mid-hold needs more than 6 months of history and an extra rule, and the next-month risk was the same (*synthetic:* 0.28 vs 0.29). It separates only over 6 months (exit 33% vs 50%).

6.4 Partial payer — "pays after reaching 4 but still rolling" (Q3)

Tested at 0.34 / 0.5 / 0.67 months paid per month since 4. Rejected:

- barely separates risk (next-month roll ~80% vs ~86%);
- unstable — ~37% leave it every month — and it **drags the stability of the fixed states from ~98% to ~91%**, because accounts flip in and out of it;
- 79% of it is "paid in the last 3 months" — the credibility axis's 2-point component.

It is one of the six natural residual clusters, so it is real behaviour — but it belongs to credibility, not here. The switch is kept in code (`partial_min_paid_rate`) so the test can be repeated (expectation E24).

6.5 Separate states for slow slide, dip-and-roll, volatile

Merged into gradual (A, B) or split once by "paid 2+ months on the way" (C). The three confuse each other in every view (shapes, clusters, trees), and the risk actions are close. Much of "volatile" (cure then relapse) is a relapse by definition.

6.6 Relapse in option A

Left out of A on purpose. Relapse is built from earlier spells — the same history credibility's 1-point "pre-delinquency clean" flag scores. In the self-test (with a clean point defined as "no earlier spell") the states explained **84%** of that point's variance. It is included in B/C because it is only 1 of credibility's 5 points and it carries more than a yes/no: *cured from bucket 3 before* is evidence of catch-up capacity. Expectation E19b measures the real overlap and decides A vs B.

6.7 "Holding or improving" clause (today's bucket ≤ 6 months ago)

Dropped. It changed **0.0%** of assignments. Sudden/gradual describe how the account fell in, so an improving account carries no misleading label without it.

6.8 Payment-based refinement of stabilized

Tested: "≤ 1 roll in 6 months AND paid 2+ months at least once" separated holders from staircases at 99% precision — but only because the synthetic generator built it that way, and it leans on payment behaviour (credibility). Replaced by the bank's own 5-of-6 rule.

6.9 Features that drift with time since bucket 4

Dropped from assignment: whole-spell roll rate, fixed-window counts (rolls in the last 6 or 12 months), months paid since 4 as a count. Straight rollers at bucket 4 and 7 looked different only because of the clock. Replaced by run-up features (fixed once past 4) and rates within the current spell.

6.10 Shape table as the assignment

Kept for discovery and stakeholder examples only. ~5–6k distinct banded shapes; the top 30 cover ~35–40% of accounts — too fragmented to assign everyone.

6.11 n-gram ranking by planted-label purity

Fixed. The first version had a meaningless "lift" (per-account presence divided by per-month code frequencies, rescaled by an arbitrary constant) and ranked by planted labels that real data does not have. Lift is now measured against each account's own codes in shuffled order: lift > 1 means the order of moves matters.

6.12 Treating +2 jumps as impossible

Replaced by repair (+2 next to a hold → straight roll) and "otherwise a roll". Real data has billing-cycle effects (example: `…3 4 3 5 4…`).

6.13 Payment-promise or inflow inputs in any state

Never used — credibility and ability own them. Their overlap is *measured* (§7).

6.14 Bust-out with up to 6 months of history

The first version allowed ≤ 6 months before the spell, chosen to match "a few months". It made `0 0 0 0 0 0 2 3 4` a bust-out — but six clean months means about four bills were paid (or nothing was due): an **early-life default**, not a customer who never paid. The synthetic 6% match did not support 6 months either: the generator's bust-out volume had been tuned to hit 6% *under that rule*, so the match was circular. Replaced by the agreed business definition: **≤ 3 months, at most one blip**. Under it the generator still gives 5.8% bust-out; every planted bust-out is caught, and 93% of the state is a planted bust-out (79% under the old 6-month rule).

6.15 Established / early-life split in option A

Not in A because, in synthetic data, it adds no risk separation (both ~94–95% exit within 6 months, by construction). It is in B/C for its action difference (life event vs underwriting signal). Real outcomes decide whether it earns its place.

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7. Overlap with the other axes (known and measured)

Overlap	Size *(synthetic)*	Treatment
holding ↔ credibility "money coming in" (2 pts)	100% of holders have money in; 44% of non-holders do too	One-directional: holding is stricter (enough to hold the bucket, 6 months). Accept and state it. E21
relapse ↔ credibility "pre-delinquency clean" (1 pt)	eta² 0.84 of the clean point explained by B states	Decides A vs B. E19b

Overlap	Size *(synthetic)*	Treatment
sudden/gradual ↔ ability	none in synthetic (by construction)	Sudden may proxy income loss on real data — fine, but say so. E20
any state ↔ locatability	none expected	E19/E20 report it

8. Real-data validation

How the backtest works from a single table

Today's string holds the past: dropping its newest *k* digits gives the string as it was *k* months ago, and the dropped digits are what happened next. So one table of latest-month strings gives the spine as of *k* months ago **and** its outcomes (next-month roll, exit to 8+ within *k*, cure, next-month state). This needs the table to **keep accounts that have since rolled to 8+ or closed** — a survivor-only table biases the outcomes (the snippet detects and warns).

Expectations (checked automatically — `dpd_states/expectations.py`)

	Check	Expected	Basis
E01	Exactly one state per account	True (FAIL otherwise)	Tree with default branch
E02	Unparseable strings	≤ 2% (FAIL)	CHAR_MAP / STRIP_CHARS set right
E03	Backtest monthly roll rates vs 70/80/90/95%	≤ 7pp off (FAIL)	Confirms orientation and reconstruction
E04	Histories starting inside the spell	≤ 10%	Full history exists
E05	Unrepaired +2 jumps	≤ 2%	Most jumps sit next to a hold
E06	Bucket mix declines 4 → 7	yes	Rising roll rates
E07	One spell only (from bucket 3)	78–92%	Bank ~85%
E08 / E09	Bust-out / holding share	4–8% / 4–9%	Bank 6% / 6%
E10 / E11	Recall of the bank's existing flags	≥ 80% / ≥ 90%	Same definitions
E12	Holding next-month roll below the fixed states	≥ 20pp gap	*Synthetic:* 40% vs ~85%
E13	Fixed states keep their state next month	≥ 95%	Run-up fixed once past 4
E14	Holding persistence next month	45–85%	Current behaviour
E15	Option A sudden share	55–80%	Most roll straight
E16	Smallest option-B state	≥ 5%	Actionable size
E17	Smaller gradual split (C)	≥ 4%	Actionable size
E18	Holding at bucket 4 vs 7	≥ 2×	90–95% roll at 6–7
E19 / E19b / E20	eta² of credibility / clean point / ability by state	≤ 0.25 / ≤ 0.50 / ≤ 0.10	Orthogonality
E21	Holders with money coming in	≥ 90%	Holding = paying

	Check	Expected	Basis
E22	Residual clusters favour B over A	ARI gain ≥ 0.15	*Synthetic:* 0.20
E23	One paid month matters (gradual at run-up ≤ 4 / ≤ 3)	≤ 0.70	Most gradual held once
E24	Partial payer stays rejected	persistence ≤ 75%	*Synthetic:* 63%
E25	INFO: exit spread across fixed states	report	Unknowable from synthetic

Every check carries an "if not met" action in the code and in the report.

Choosing the option on real data

1. Any **FAIL** → fix config/data first; read nothing else.
2. **E19b high** (relapse ≈ credibility clean point) **and** the risk team does not want the cured-before nuance → **A**. Otherwise **B**.
3. **E17 passes and** the risk team has distinct actions for slide vs fought-back → **C**.
4. **E25 ≥ 5pp** → the mechanism carries risk on its own; report it to the risk team as a bonus.

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9. Validation done so far *(synthetic)*

- **Calibrated generator** (`dpd_states/synth.py`): eight planted journey patterns (bust-out, clean abrupt shock, slow slide, staircase, stabilized, repeat episodes, dip-and-roll, volatile); full lifecycles to exit; length-biased snapshot like a real spine; payment holidays (frozen buckets) and billing-cycle +2 jumps; one continuous tenure distribution. Matches the bank's figures: roll rates 70/80/89/95% (target 70/80/90/95%), bust-out 5.8% (6%), stabilized 6.5% (6%), single spell 83% from bucket 3 (85%).
- **Method report:** `validation/dpd_states_validation_*.md` (`python -m dpd_states.run_synthetic`).
- **Real-data path self-test:** `validation/selftest_report_*.md` (`python -m dpd_states.selftest`) runs the exact analysis and expectations the workplace run uses, on synthetic data with fake axis columns. All checks pass except two expected WARNs: E19b (relapse ≈ credibility clean point, §6.6) and E16 (sudden_early_life 3.4% under the 12-month established cut — a property of the synthetic tenure distribution, §5).

What synthetic data cannot tell you: whether the rolling states differ in outcome (they share post-4 dynamics by construction), how big the real overlaps with credibility and ability are, and how much payment holidays contaminate holding. The real-data run answers all three.

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10. Open decisions

1. Option A, B or C — decide from E19b, E17 and the risk team's actions (§8).
2. Spell definition (from bucket 3 or 4) — E07 shows which matches the bank's 85%.
3. Bust-out history threshold — confirm 6 months against the bank's existing rule (E10).
4. Payment holidays / restructurings — if flags exist, exclude frozen months before Q2.

Glossary

Bucket — DPD band, one digit per month · Roll — +1 bucket, nothing paid · Hold — same bucket, one month's due paid · Cure — back to 0/1 · Spell — continuous months at ≥ 2 · Blip — 1–2 month spell never past 2 · Run-up — last clean month to first month at 4 · Spine — accounts currently at 4–7 · eta^2 — share of an axis score's variance explained by the states (0 = orthogonal) · ARI — agreement between two groupings (0 = chance, 1 = identical).